

Intro to Cog Sci

What is Cognition? The action of perceiving things.

Quizzes 20%

Mid Sem 20%

End Sem 30%

In Class Activities 5%

Expts - 5% bonus + reflection
paper

Processing is top down. We are influenced by our experiences, memories.

Cognition processes become stable & then we "see"

Usually stable, sometimes bistable

We usually predict what we are about to perceive before actually perceiving it.

How do we rapidly process & interpret information?

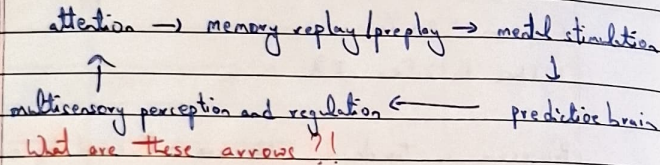
Prior experiences and perceived/observed stimuli

~100B neurons → 160T connections in the brain.

Predictive Brain

Not so prevalent in babies, but over time, the past experiences cause us to predict.

But when predictions don't match with our perception — Surprised — learning experience



Neurons fire repeatedly, & no. of pulses and no. of regions determine the cognition.

↳ P, Y, S, & waves

↳ active
 ↳ sleeping

Behaviourism & Cognitivism

Programs run on symbols; neural computation uses mental representations.

↳ The specific config of neurons that fire in activity patterns for any object/stimulus etc.

↳ encodes features of the object. Any object is defined by its features. Same for abstract concepts

Noisy representation space causes confusions. People can train their brains/minds to be more sensitive to certain feature differences

Marvin's Three Levels of Analysis (of a system)

- (i) Computational :- What is the problem? What is the I/O?
- (ii) Representational :- What is the procedure?
- (iii) Implementational :- How is the procedure implemented?

Embodied perception:- How do we perceive while interacting with the environment. Also embodied cognition. Surroundings become part of the representation.

Intro to Sensation and Perception

100 bil neurons → 160 tril connections [we just wrote this last class orghh!]

Sensation :- How we felt feel, inputs from environment. Mechanistic process, sensory mechanism, signal processing

Gather information from the environment via the senses.

Conversion of all stimuli into uniform signals.
Not interpreted.

Perception:- Processing & understanding these signals.

Central Nervous System:- Brain & Spinal Cord.
Primary cognition

Peripheral Nervous System:- Relay to the CNS,
only small, simple processing loops.

Relay Neurons:- High fidelity for transmitting info.

Neurons are unidirectional

Parts of a neuron:-

- ↳ Cell body
- ↳ Dendrite → lots of branches → dendritic tree
 - ↳ Connects neurons together
 - ↳ Learn to build contacts
- ↳ Axon → ion carrying channels (Na^+ & K^+)
 - ↳ This carries action potential
- ↳ Axon Terminals → contain synaptic vesicles
- ↳ Myelin Sheaths → insulation for the axon → made of fats

Neurotransmitters - Chemical messengers that takes messages

across neurons → Chemical Synapses

Electrical Synapses → Neurons are physically connected

Info always flows from Axon Terminals to Dendrites

Jns b/w myelin sheaths act as signal repeaters to increase the strength → happens very rapidly

↳ Nodes of Ranvier

Synaptic Cleft:- Small, microscopic gap b/w neuronal connections

Ant, Type, duration of neurotransmitters allow for various signals to be sent. Hence the synapse

When a neuron is at rest, there is a potential difference in charge inside & outside the neuron (innards are -ve since there are fewer +ve ions than outside)

When the neuron gets neurotransmitters, it opens channels near the dendrites & cell body to let the Na^+ ions in.
⇒ a threshold of firing, resting voltage = -70 mV

The resulting signal is the action potential
Current is in μA

Cells either fire or they don't (ReLU style)

Axon hillock is where the decision to fire is taken
↳ in bio cell body & axon.

No. of axon terminals ≥ 1

Receptors:- Every neurotransmitter has a specific site for binding.

Recreational drug drugs fool the ^{receptors} to mimic certain neurotransmitters.

These neurotransmitters release proteins (??) which causes the cell body to open the channels.

Reuptake:- Degradation of neurotransmitters. Either the parent cell takes it away, or they are destroyed/absorbed by other cells.

* Thermoreceptors, Photoreceptors, Chemoreceptors, Mechanoreceptors
↳ taste & smell
↳ Nociceptors (Pain)

Perception is the organising and interpreting to create meaningful experiences of the detected physical energy by

sensory receptors (sensation)

Brain $\in$ Frontal lobe, Parietal lobe, Occipital lobe, Temporal lobe, Cerebellum & Medulla

Top down & bottom up processing can happen

Top down can dominate bottom up by choice, vice versa only when the signal itself is very strong

Vision:-

We have a dominant visual cortex (largest compared to other sensory areas). We are visual creatures

We need

↳ Light

↳ Eyes $\leftarrow$ limited to the visual spectrum (400-700 nm)

↳ Brain

We have separate cones for red, blue & green.

$\lambda \rightarrow$ shade/colour

amp $\rightarrow$ brightness

purity $\rightarrow$ how many colours can we perceive.

Light enters through the pupil. Iris controls the size of the

pupil

light enters through the cornea, then through the pupil (controlled by the iris). It then enters the lens which it bends the rays & focuses the light. The light then focuses on the retina which has light sensitive tissue (rods & cones).

The image is inverted at the retina.

Blindspot where the optic nerves ~~are~~ ^{exit the eye} due to a lack of photoreceptors

Myopia:- Near sightedness - Can't ^{focus on} see far objects

Hyperopia:- Far sightedness - Can't focus on near objects

Astigmatism:- Distortedness of image

Phototransduction $\leftarrow$ perceiving light.

Fovea:- Image is clearest here. Peripheral vision has lower fidelity.

Both eyes convey ^{information} hemispheres to both hemispheres - Such that all info about the left visual field goes to the right hemisphere & vice versa.

The place where the nerves cross over is called the optic chiasm

Cones always needs rods to be active. Have only rods activate in dim light / darkness.

Rods / Cones $\rightarrow$ Bipolar Cells $\rightarrow$ Ganglion Cells $\rightarrow$ Optic Nerve

approximators

Lateral Geniculate Nuclei:- Parts of our brain act as gating mechanism, only let in that which we have attended to.

If the optic chiasm is damaged, then you the individual might go blind, or lose peripheral vision

Colour Perception:-

Trichromatic Theory:- Three cones $\rightarrow$ three colours
Explains colour blindness.

Opponent Colour Theory:- Pairs of colours inhibit each other. When one is removed, the other dominates

This theory explains the afterimage.

Parietal lobe:- Where have you seen? - Dorsal Stream

Temporal lobe:- What have you seen? - Ventral Stream.

Perceptual Constancy:- Even as aspects of signals change, the perception remains constant.

Perceiving Form:- Gestalt Perception

What we perceive $\geq$ features observed

We see wholes, we do not perceive visual stimuli featurewise.

Grouped by 6 features (ref slides)

We can specifically attend to what we believe to be the central figure - can't do this if we look only at features.

Depth

We can perceive depth because of binocular vision.
→ evolutionary advantage.

monocular cues - shadow, texture, gradient, occlusion, linear perspective etc.

Motion:- Presence & absence of certain signals causes us to perceive motion.

The features are held relatively constant to be able to perceive stably.

Activity on 24th Aug

Senses are not compensatory, they are competing in the brain.

Inattentional Blindness:- A failure to perceive objects that are not the focus of attention.

Visual Agnosia:- Can't perceive objects but can identify shapes.

Motion Blindness:- Inability to perceive moving objects.

We can always perceive faces regardless of variation because we have been trained through our life.

Object Perception:-

Basic features - Geons - are unified by the brain to form a whole.

Biederman's Theory:- Each item can be broken down into Geometric Items → Geons.

Multiple Vision lectures ahead

Somatosensation:-

Thalamus is the relay centre of the brain. Olfactory signals bypass this & directly reach the Olfactory cortex/nbbs.

→ Touch, Temp, pressure, pain etc.

↳ Free Nerve Endings ← Nociceptors

↳ Basket Cell Fibres ← Pressure, heat, etc. ← Mechanoreceptors, Thermoreceptors, Proprioceptors

Why switch sides? What about layered stimuli - touch.

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To show intensity $\uparrow$. After a point, it just keeps firing due to tissue damage.

Homunculus - Topological Map of the body in the brain.
Greater sensitivity $\rightarrow$ Greater representation

Tactile Perception:-

Tactile Acuity $\leftarrow$ Two point discrimination $\leftarrow$ minimum distance b/w two pressure to distinguish.

Dependent on the receptive field.

Plasticity

- $\hookrightarrow$ Structural $\leftarrow$ Connection changes; myelin sheaths are better
- $\hookrightarrow$ Functional

Pain:- Unpleasant emotional / sensory experience associated with tissue damage

- $\hookrightarrow$ Nociceptive
- $\hookrightarrow$ Neuropathic
- $\hookrightarrow$ Nociceptive

Gate Control Theory of Pain:- This is why we apply pressure / heat

- $\hookrightarrow$ Pain perception is controlled by gates

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$\rightarrow$ diameter $\leftarrow$

Small fibres open gates (pain), large ones close gates

Gestalt - 1 :- Proximity / Similarity

$\hookrightarrow$ Principles to group info together to improve processing.

Auditory Stimulus has no boundaries. Just a continuous series of changes in pressure.

$\rightarrow$ lower mechanism to amplify.

Sound $\rightarrow$ Eardrum $\rightarrow$ Ossicles (3 bones in middle ear)
 $\rightarrow$ Cochlea (inner ear)

Outer Ear - Pinna - Funnel Sound

Middle Ear - Amplifier

Inner Ear:-

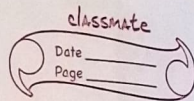
$\hookrightarrow$ Vestibular System for balance (semicircular canals - filled with fluid)

$\hookrightarrow$ Cochlea - converts sound pressure into e- signals.
Has lining of fine hair called cilia. that moves when vibrated & send e- signals / nerve impulses

Higher end of the cochlea is exposed to every sound. because it is compromised $\rightarrow$ loss of ability to perceive high V.
Cilia are on Basilar Membrane.

Organ of Corti contains the receptor to transduce signals into nerve impulses

Tonotopic Map



Theories of Hearing:

(i) Place Theory Coding Theory - encoded at diff places in the cochlea
But neurons have a max firing V

(ii) Temporal Coding Theory - encoded in the timing of neural firing across neurons

(iii) Volley Theory - sync of several adjacent neurons.

Ref Slides

Primary Auditory Cortex:- Heschl's Gyrus

Secondary Auditory Cortex:- Home to Wernicke's Area.

Wernicke's is split across both hemispheres.

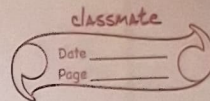
Left part of the auditory cortex processes with ↑ temporal resolution.

Auditory has a what and a where pathway
↳ anterior ventral ↳ posterior dorsal

Broca's - Speech production / expression

Arcuate Fasciculus - white matter bundle connecting
• Broca's & Wernicke

"Sounds Like a Winner"



Melodic Intonation Therapy - Sing instead of speak.

↳ learns a tune online while speaking

↳ not pre-made tunes

Sound & Freq → perceived as pitch

& amplitude → perceived as loudness

& timbre → how the different overtones / harmonics (not a straightforward mapping)

↳ helps us differentiate sounds that have the others fixed.

amplitude:- Objective measurements of the degree of change in atmospheric pressure.

Loudness is the psychophysiological correlate

frequency:- No. of sound wave cycles perceived

Pitch is the psychophysiological correlate perceived

The Mel Scale:- Perceptual scale of equidistant pitches

Pitch of a periodic tone = pitch of the sine tone equal to the first period.

Virtual pitch vs Spectral pitch

Timbre / tām̃bāṣ/

↳ No. & relative positions of harmonics

↳ tone-colour

Psychoacoustics with basket $\rightarrow$ that which is not pitch, amplitude & duration.

Cocktail Party Effect:-

$\hookrightarrow$ focusing on one specific auditory stimulus while filtering out the others $\rightarrow$ selective attention to specific timbres

Necessary for survival in the wild.

Auditory Hallucinations:-

Perceiving sounds w/o auditory stimulus.

- $\hookrightarrow$ Abnormal activation of the normal auditory, long perception & production pathways.
- $\hookrightarrow$ activations of Primary Auditory Cortex, amygdala, hippocampus, frontal & sensorimotor cortices & Broca's area
- $\hookrightarrow$ sensory deprivation (deprivation?)
- $\hookrightarrow$ deprivation of social interaction
- $\hookrightarrow$ heightened emotional states

~~Ben~~ Tinnitus!

Hearing Loss:-

- $\hookrightarrow$ Conductive $\rightarrow$ Outer & Middle
- $\hookrightarrow$ Sensorineural $\rightarrow$ Inner & Neurons
- $\hookrightarrow$ Mixed Hearing

Olfaction:-

Sense of smell

- $\hookrightarrow$ Identifying toxins

Orthonasal $\leftarrow$ Through the nose

Retronasal $\leftarrow$ Through the back of the mouth

Bypasses the thalamus $\rightarrow$ goes to olfactory bulb in the fore brain.

Glomeruli $\leftarrow$ olfactory chemoreceptors

- $\hookrightarrow$ in the olfactory bulb, maps all smells to specific firing patterns of glomeruli.

Gustation

Results from firing patterns of taste buds $\rightarrow$ have a lot of Taste Receptor Cells.

Labelled-line model:-

Receptor cells are tuned to a single taste.

Across Fibre↳ ~~diff. cells~~

↳ one cell → multiple qualities

and/or

↳ ~~one~~ afferent fibre carries multiple signalsRef slidesMultisensory Perception

Sensory events are never experienced in isolation

Convergence Zones ← Where multisensory ^{integration} percept occurs

↳ Superior Colliculus

Super / Supradditivity :- multisensory neuronal firing > firing for individual stimulusSubadditivity :- vice versaPencil time

Sonic Chip expt:- Judge freshness & crispness of chips while either amplifying or dampening the sound of the chip being eaten

Phonetic value of sounds transfers to the odours irrespective of the value of the odour ~~for~~ itself.Orbitofrontal Cortex - (OFC)

↳ receives inputs from multiple cortices

Synesthesia:- An exp in one sensory modality seems to induce an exp in another non-stimulated sensory modality.
↳ increased conn. b/w sensory areas → involuntary and sustainedAdjacency Principle:- It is a result of the neural areas processing the information are adjacent.
Not a complete explanation

Learned associations for cross modal sensory perception

Sensation, Perception & Cognition
Detection, Interpretation & Meaning making.

Existing knowledge guides this dynamic process

Localisation is Distributed processing

↓

↳ spread across multiple areas

Spl areas for spl stimuli

Neural Code - patterns of firing of neurons/neuronal populations for specific objects/stimuli

Distributed Processing vs Distributed Coding



Specific features processed
at specific areas



How do neurons encode
objects/features

Feature detectors - certain neuronal populations respond to specific stimuli/features

Brains try to achieve efficiency in coding - good prediction machine - minimise prediction error, uses top-down & bottom-up processing

Rapid, Effortless, Adaptive, Selective, Controlled by Patterns, involves reasoning, dynamic - nature of perception

Gestalt Principles of Organisation

↳ Shape/Form

Pickel Face Effect → Perceive continuity even in discontinuity

Whole >>> > parts

Brain tries to perceive things as wholes than as parts.

(i) Similarity

(ii) Continuation - Pickel Face Effects

(iii) Proximity

(iv) Closure

(v) Figure Ground ← depends on past knowledge

(vi) Common Fate

But these laws are heuristic → can lead to errors

Perception is shaped by knowledge of regularities in the environment

Bottom Up → What do I see → begins with stimulus → integration

Top Down → What I seen this before → begins with prior knowledge
→ Try interpreting the stimuli.

Models of recognition

↳ Feature Matching

↳ Reassembly-components

↳ Template Matching

↳ Configurals

Geons - primary building blocks of perceived objects

↳ Form / for recognition by components

Relation b/w Geons is necessary for recognition.

Damage to fusiform area → loss of ability to perceive objects, can only process lower order geons.

Higher level areas prime lower level ones

Perceptual Constancy:-

Size & Shape for visual systems

Timbre

Models of Recognition

Dorsal stream - Action pathway

Ventral stream - Perception pathway

Single dissociation → lesion in an area disrupts one aspect of, not others

Double dissociation →

Paul Thagard

Actions are understood by simulating them covertly in the mind - the mental representation of the actions are triggered.

Bapi Time! (Is that good or is that bad future look?)

* Vishnu's Slides? Two T

5-6 lectures? :sob:

Bring back Vinna please

Memory:- Everything we experience. Procedural, Semantic, Episodic etc.

↳ Deals with stuff like recall, reproduction

Representation:- An entity that allows us to use information in a flexible way.

You can't have representation w/o memory & vice-versa.

Computational Theory of the Mind:-

Brain is the hardware, mind is the software.

Mind (problem solving) ← representations + cognitive processes

↳ performs steps involving representations

Representations used as stand-ins for objects

Given an object, when an animal does actions related to the object, then the representation undergoes transformation.

Representations allow us to anticipate

Anticipation presupposes the existence of a representation.

Reactions or reflexive actions have only implicit representations that are not flexibly used. Reactions bypass the representational layer.

Connectionism:- Resulted in distributed coding.
A grandmother neuron.

Neural Code - firing patterns of neurons that stand for objects.

Inferior Temporal Lobe:- Object representations.

Downstream neurons decode these codes.

Neuroscience studies - measures correlation b/w codes and the external world.

Do these representations participate in meaningful activities with the object?

The algorithms from Marr's theory act on representations.

This approach is more about internal computational framework and the role of representation.

Representation:-

- (i) Correspondence:- Is a correlation
- (ii) ^{Causation} Correlation:- Must have a functional role
- (iii) Teleological Context:- Must be purposive

It has semantic context

Neurons have response tuning curves $\rightarrow$ a set of neurons can now code multiple stimuli

To measure causation $\rightarrow$ deactivate an area and measure behaviour.

Examples: TMS, drugs, burn it, freeze it etc.

We can also increase activity in certain areas.

Representational Similarity Analysis

Hippocampus represents space & time.

Or is it a sequence generator?

Decision Making and Cognitive Biases

Decision Making Environment:

- (i) **Certainty**:- Outcome is known
- (ii) **Risk**:- Outcome is uncertain, but the probabilities are known
- (iii) **Uncertainty**:- Neither outcome nor probabilities are known

Rational Decisions:-

Informal decision making

Two systems of decision making

Heuristic

- (i) **System -1**:- fast, ~~and~~ automated, immediate, less effort, exp dependent

Rational

- (ii) **System -2**:- slow, deliberated, weighing options, conscious information dependent

We have specific brain areas for both systems.

Sometimes system-1 decisions never reach awareness.

Heuristic - mental shortcut

Bias - outcome / effect produced by Heuristic processes.

Can lead to beliefs / stereotypes.

- (i) **Availability Bias**:- How easily do examples rise to our mind when we think about it.
- (ii) **Confirmation Bias**:- Selectively seek & interpret information that supports pre-existing beliefs / preferences.

Thinking Fast, Thinking Slow.

- (iii) **Representation Bias**:- Judging something based on how closely it resembles a stereotype.
- (iv) **Anchoring Bias**:- Tendency to excessively rely on an initial piece of information while making estimates.
- (v) **Hindsight Bias**:- Post hoc reasoning / justification for decisions made.

Framing Effect:- The way options / alternatives are framed influences decisions made.

Loss Aversion:- The possible losses look more than the prospective gains.

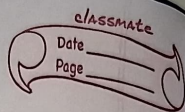
People evaluate outcomes relative to a reference point (expectation) rather than simply according to their absolute values.

Cognitive Bias:- Systematic deviation from ^{normative reasoning in individual and decision making} rationality. Repetition results in formation of heuristic processes to reduce cognitive load.

Primates are Mammals with complex brains can show some level of system-2 thinking (chimps, gorillas, orangutans).

Two lectures by Sourabh Todoraya
Gotta catch 'em all.

Can you have thoughts that you are not cognisant/conscious of?



Embodied Cognition:- $\hookrightarrow$ But we're conscious of absence of dreams

Why?

Philosophy of Mind: No intention/volition in the brain, it is all in the mind.

epistemic status $\rightarrow$ propositional in nature.

Mental Content can be conceptual & non-conceptual

World is made up of facts (through propositional knowledge)

Embodied cognition disagrees with this

We need by knowledge to use propositional knowledge.

Cartesian Dualism:-

Descartes $\rightarrow$ thoughts are entry points to consciousness.

Mind is conscious, immaterial & indivisible

Body is extended, material & divisible

Physicalism:- All mental events ~~are~~ are caused by the brain.

Type Identity Theory:- Mental states correspond to brain state types.

UT Plus $\rightarrow$ differentiable b/w "is" of identity & "is" of predication.

Consciousness is a brain process $\rightarrow$ Water is H_2O not Water is blue

Mental events are empirical descriptions & brain process is the accurate, scientific description of the same process

Criticism of Physicalism - Multiple Realisability Hypothesis

We can imagine people with neural archs not like ours having emotions etc like us $\rightarrow$ this is a logically possible
Consciousness is a brain process is a contingent identity

Behaviourism:- Only evidence we have of mental states are the outward observable behaviours.

Ascribing a mental state = ascribing a behavioural disposition to that person.

$\hookrightarrow$ behavioural dispositions = mental state

The mind is just a collection of stimulus-response mappings.

Mind is an independent entity = category mistake

Counter:- We can imagine super-spiders who do not show pain even while in pain. These behavioural dispositions are also contingent conditions, not necessary for mental states.

Self is the narrative centre of gravity

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Beliefs:- Impossible to have any disposition unless we believe in that. External world causes beliefs

Network of beliefs $\rightarrow$ rise to behaviour.

$\hookrightarrow$ causal & coherent relationships

Functionalism:- Mind as a machine. We do not care about what constitutes ^{the} ~~the~~ functionality.

Environment
Input:- Sensory Input + Belief

Output:- Behaviour

Consciousness = Rational Decision Making?

Thoughts are the only way we can gauge our own consciousness.

Take the thought experiment of having a brain with V6 but w/o experiencing. Can this brain be conscious?

Intentionality:-

Feature of consciousness that
Consciousness is always of something, internal or external;
also known as object-directedness

This is a mark of the mental.

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Intentional states require background knowledge, capacities & practices.

Background beliefs are a result of ^{expectations shaped by} prior experiences.

For any intention to be functional, we need background knowledge.

Knowledge = Justified True Beliefs.

You can't add new information through propositions

Hume argued that induction is not explicable

Any knowledge claim requires background knowledge

Even primitive fears like falling need to be realised, consciousness does not exist in the head.

Knowing That:- Knowledge of the world. Propositional Knowledge

Knowing How:- Ability to do actions. Experiential/Capacity.

Initially body is not habituated to actions, explicit cognition is necessary.

Over time, habituation $\rightarrow$ usage of embodied cognition / consciousness.

Automation is only possible with embodied knowledge.

Pre theoretical knowledge $\rightarrow$ background knowledge

$\Downarrow$

normative in nature. Limited by the embodied capabilities.

Objects as affordance

Perception of a physical object requires :-

$\hookrightarrow$ an object $\hookrightarrow$ a knower.

We never perceive objects fully $\rightarrow$ but we trust in the wholeness/completeness (Gestalt Principles) because this is afforded by our embodied capabilities.

Based on our kinesthetic abilities. This is pre theoretical & background.

Perception arises from bodily interaction with the environment.

Mere sensory input is insufficient, active sensorimotor engagement is essential.

Perception is not representational throughout.

Background Knowledge is not blind, but is pre-reflective; shaped by embodied, embedded, extended & enacted.

Consciousness can be: positional (single object-focus) (has an anchor) or non-positional (background awareness).